

Morbidity & Mortality Conference Case Presentation

Date of Conference: January 15, 2023

Presenter: Dr. J. Mills, Hematology Fellow

Case Summary:

We present the case of Ms. Pamela Evans (born February 12, 1950, MRN: AML071), a 72-year-old female with a complex medical history who experienced a rapid and fulminant course of adverse-risk Acute Myeloid Leukemia (AML). The case highlights the significant challenges in managing therapy-related myeloid neoplasms (t-MN) or AML with myelodysplasia-related changes (AML-MRC), particularly in the elderly.

Initial History:

Ms. Evans was referred to our hematology service in February 2022 for persistent, symptomatic anemia. Her past medical history was notable for:

- Stage IIB Breast Cancer diagnosed in 2012, treated with lumpectomy, adjuvant chemotherapy (Doxorubicin/Cyclophosphamide followed by Paclitaxel), and radiation. She was considered cured.
- Osteoarthritis, severe, requiring chronic NSAID use.
- Chronic Kidney Disease Stage IIIa (baseline Creatinine ~1.3 mg/dL).
- Paroxysmal Atrial Fibrillation, on Metoprolol.

For six months prior to her referral, she had noted progressive fatigue, dyspnea on exertion, and the appearance of petechiae on her lower extremities. Her primary care physician had been monitoring a slow decline in all her cell lines.

Diagnostic Workup (March 2022):

An urgent bone marrow biopsy was performed on March 4, 2022.

- **Aspirate Smear:** Revealed 45% blasts of myeloid origin, with prominent dysplastic features in the erythroid and megakaryocytic lineages.
- **Flow Cytometry:** Confirmed an aberrant blast population positive for CD34, CD117, CD13, CD33, and myeloperoxidase (MPO).
- **Cytogenetics:** This was the critical finding. The karyotype returned as **45,XX,del(5q)/46,XX**. The deletion of the long arm of chromosome 5 is a classic high-risk feature, often associated with prior toxic exposures (i.e., chemotherapy).
- **Molecular Studies (NGS Panel):** Negative for common mutations such as FLT3, NPM1, IDH1/2. TP53 was wild-type.

This constellation of findings—history of cytotoxic therapy, antecedent cytopenias with dysplasia, and del(5q)—firmly placed her in the **ELN 2022 Adverse-risk** category, consistent with a therapy-related AML.

Treatment Course & Rationale:

A comprehensive goals-of-care discussion was held with Ms. Evans and her family. Given her age (72), borderline performance status (ECOG 2), and underlying CKD, she was not a candidate for intensive induction chemotherapy (e.g., "7+3"). The standard of care for this population is a hypomethylating agent (HMA). We initiated

treatment with **Decitabine** (20 mg/m² IV daily for 5 days on a 28-day cycle), starting on March 14, 2022.

- **Cycles 1-3 (March - May 2022):** She tolerated the initial cycles reasonably well. Her transfusion needs for packed red blood cells decreased from weekly to bi-monthly. A repeat bone marrow biopsy after Cycle 3 showed a reduction in blasts to 15%, indicating a partial response, but with persistent disease.
- **Cycles 4-6 (June - August 2022):** Her response plateaued. She remained transfusion-dependent for both platelets and red cells. She began to experience worsening fatigue and recurrent admissions for neutropenic fever, despite a lack of identifiable infection. This suggested her underlying disease was becoming more aggressive and resistant.

On September 10, 2022, she was admitted with profound pancytopenia (ANC < 100, Platelets < 10k) and bleeding gums. Her peripheral blood smear showed 30% circulating blasts.

Terminal Decline:

The decision was made not to proceed with the planned next cycle of Decitabine. We discussed second-line options, including Venetoclax-based combinations or enrollment in a clinical trial. However, Ms. Evans' clinical status deteriorated rapidly. She developed hospital-acquired pneumonia and became increasingly frail and somnolent. The family, honoring her previously stated wishes, opted to transition to comfort care. She was transferred to our inpatient hospice unit on October 12, 2022. She passed away peacefully nine days later, on **October 21, 2022**.

Learning Points & Discussion:

1. **Prognostic Power of Cytogenetics:** This case underscores how adverse cytogenetics like del(5q) drive prognosis and often predict poor response to standard therapies like HMAs.
2. **Limited Efficacy of HMAs:** While HMAs are the standard of care for this patient population, the duration of response can be short, particularly in t-MN. Ms. Evans' 7-month progression-free survival is unfortunately typical for this disease biology.
3. **Importance of Early Palliative Care:** The early integration of palliative care principles, even while pursuing disease-directed therapy, is crucial in high-risk AML to prepare patients and families for the high likelihood of treatment failure.

This case serves as a sobering reminder of the aggressive nature of therapy-related myeloid neoplasms and the urgent need for more effective, novel therapeutic strategies for this vulnerable patient population.